SCHOOL OF ENGINEERING

DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

Computer Science and Engineering (AIML)

Minor Project Phase-I

"MEDICART: DESIGN AND IMPLEMENTATION OF AN ONLINE MEDICINE STORE SYSTEM"

Under the Supervision
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TITLE: MEDICART: Design and Implementation of an Online Medicine Store System

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Introduction

MEDICART is an online medicine store that helps people buy medicines easily through the internet. Instead of going to a pharmacy, users can search for medicines, upload prescriptions, and order them from home.

This system saves time and effort, especially for sick people or those living far from medical stores. It also provides quick and safe delivery of medicines.

Overall, MEDICART makes the process of buying medicines simple, fast, and convenient.

Literature Review

Paper Title	Journal Name and year	Technology / Design	Metrics	Advantages	Limitation
E-Pharmacy User Acquisition Systems	Healthcare Digital Forum, 2021	Web-based E-commerce Platforms	User retention registration steps	Improves onboarding	High drop-off rate if process is lengthy
Netmeds / 1mg System Architecture Study	Industry Case Study, 2022	Micro service Architect - ure	Scalability, response time	Highly scalable and modular design	Complex implementati -on and maintenance
REST API Based Web Architecture	Express.js Docs, 2022	RESTful API Design	Response time, modularity	Clean separation	Needs proper security hand
Redux State Management in Web Apps	Redux Documentation, 2022	Local Storage Persistence	State consistency, session	Maintains user session effectively	Adds complexity to frontend

Problem Statement

Despite the proliferation of e-commerce, many local pharmacies still operate on manual systems, leading to:

- **Inaccessibility**: Limited operating hours and physical distance.
- **Information Gap**: Difficulty in finding detailed information about specific medicines.
- **Queue Management**: Long waiting times for customers in physical stores.
- **Inventory Errors**: Manual tracking of stock levels and expiry dates is prone to human error.
- **Payment Security**: Traditional cash transactions can be less secure and harder to track for large orders.

The MediCart system addresses these challenges by automating the entire process from browsing to delivery.

Aim & Objective

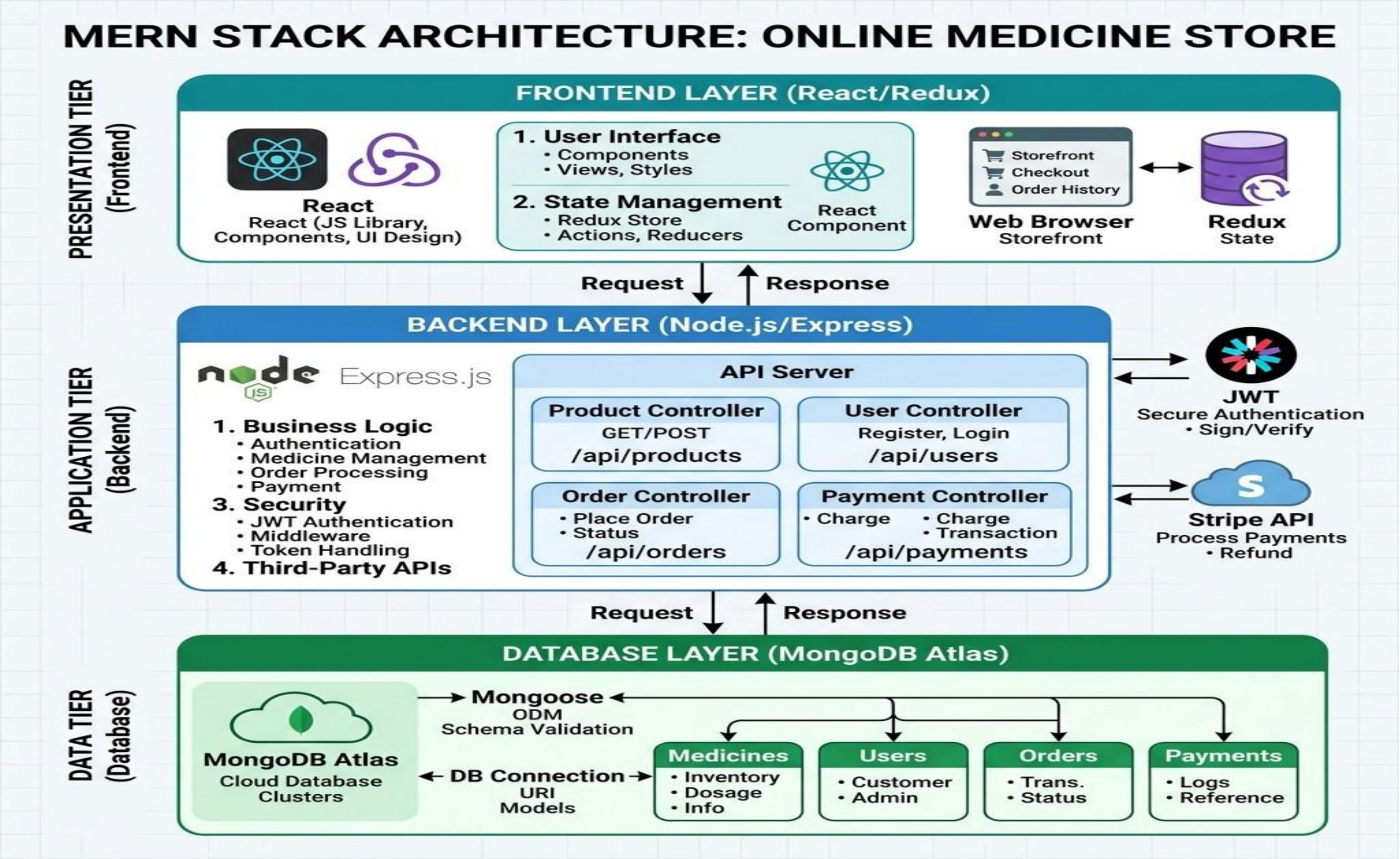
Aim: To design and develop an efficient and user-friendly online medicine store called MEDICART that enables users to search, order, and receive medicines conveniently from their homes, while ensuring accuracy, reliability, and timely delivery.

Objective:

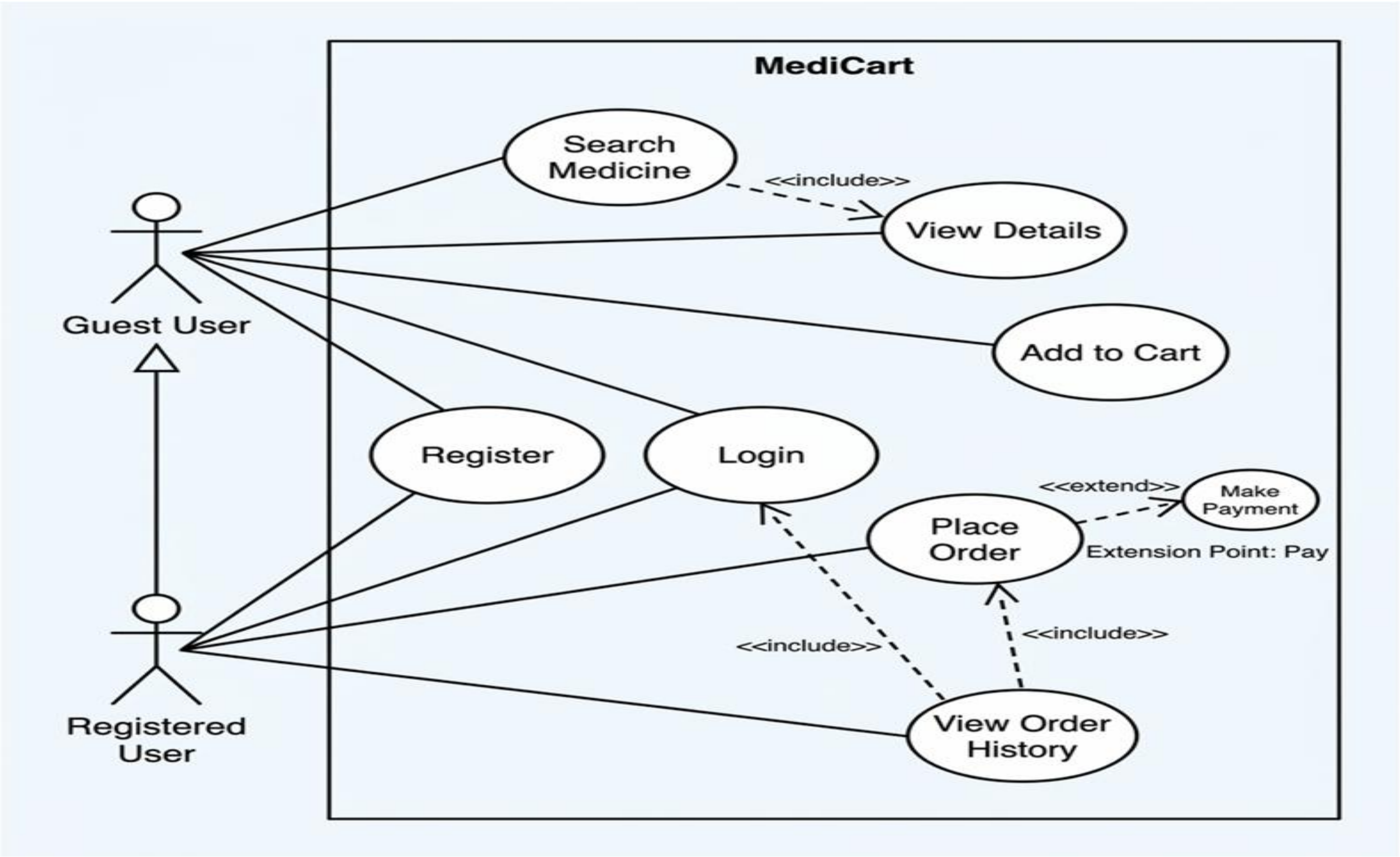
- To develop a secure and scalable platform for purchasing medicines online.
- To integrate a secure payment gateway for multiple payment modes.
- To simplify the shopping experience through an intuitive UI/UX design.
- To enable efficient inventory management via a centralized database

Methodology: The structured approach, techniques, and processes used to design and develop the MEDICART system, including requirement analysis, system design, implementation, testing for the project objectives effectively.

Model Architecture in detail



Methodology:



Results and Discussion:

Results: The MediCart system was implemented successfully with all required features working properly.

- The **Home Page** displays medicines with title, price, and description clearly.
- The **Product Detail Page** shows complete information such as usage, side effects, and availability.
- The **Shopping Cart** allows users to add items, update quantity, and view total cost accurately.
- The **Login and Registration** system works with real-time validation.
- The **Checkout and Payment** process is completed securely without errors.

Discussion: The MediCart system shows good performance, usability, and security based on testing.

- The **user interface** is simple and easy to use, helping users navigate without confusion.
- The use of modern technologies improves overall system speed and responsiveness.
- **Database indexing** ensures fast searching and efficient data handling.
- The system can support **multiple users simultaneously**, making it scalable.
- Security measures like password hashing protect user data effectively.
- Efficient API design reduces load on the server and improves performance.

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